On-Policy Distillation on HyperPod — code to eval dashboard

Qwen3.5-9B teacher → Qwen3.5-2B student (LoRA r=128) · GSM8K · no SFT · one ml.g5.24xlarge · driven from one notebook

SETUP — ONCE

1 Code
Infra template, training & eval code, notebook

2 Provision
CloudFormation: EKS, HyperPod, FSx, S3

3 Configure
Code & weights to S3, one ConfigMap

BASELINE — ONCE

4 Baseline
Serve the 9B; eval 9B · 2B · 0.8B
94.3 · 74.8 · 52.0%
→ pick the 2B

RUN — EACH EXPERIMENT

5 Distill
On-policy loop, 2B LoRA student, no warm-up

6 Evaluate
GSM8K: after, greedy + 8-vote majority, vs. the baseline

7 Dashboard
74.8% → 89.2%
8-vote majority (84.0% greedy)
teacher 94.3%

Inside step 5 — the on-policy loop

The student learns from the teacher's feedback on its own outputs

Student samples
64 prompts × 4 answers

→ **Teacher scores**
every token of those answers

→ **Student updates**
reverse KL · answer masked

repeat — the updated adapter is hot-swapped into the sampler every step

One node, four GPUs

Each GPU belongs to exactly one job

GPU 0 Teacher	GPU 1 Teacher
GPU 2 Student trainer	GPU 3 Student sampler + eval

/fsx FSx for Lustre — checkpoints, metrics, results
/s3 S3 — code and model weights